



# PTT Global Chemical Public Company Limited

## GC Management System and Process Safety

**P-(Q-MP)-OEMS-005**

**Hazard and Operability Study (HAZOP)**

**Created by :** Mr. Pongpasin Tanaruangarmorn

Senior Safety Engineer

**Approved by :** Mr. Warakorn Decha

Vice President

### Reviewer list

| Reviewer             | Position         | Unit Code |
|----------------------|------------------|-----------|
| Mr. Noraphol Sookkho | Division Manager | Q-MP-TS   |

### Edition records

| Rev. | Effective Date | Detail                                                   | Updated by                       |
|------|----------------|----------------------------------------------------------|----------------------------------|
| 1    | 21/03/2025     | Create New Document                                      | Miss Chaowarat Bunditkamol       |
| 2    | 23/06/2025     | เปลี่ยนรายชื่อผู้เขียนให้ถูกต้องตามหน่วยงาน              | Mr. Pongpasin<br>Tanaruangarmorn |
| 3    | 28/01/2026     | Revise to add new KPI and detail relate to PSI readiness | Mr. Pongpasin<br>Tanaruangarmorn |
| 4    | 26/05/2026     | ตรวจสอบเอกสาร                                            | Mr. Pongpasin<br>Tanaruangarmorn |

### Related Units

| Unit Code | Unit Name                               |
|-----------|-----------------------------------------|
| Q-MP      | GC Management System and Process Safety |
| Q-MP-TS   | Technical Safety Service                |

### Related KPI

| KPI Measure                                  | Description / Calculation                                                                                                                                                                                                                                               | Target (unit) |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| HAZOP-1: PSE Tier 1, 2 relate to HAZOP study | Number of PSE (Tier 1 and Tier 2) for MoC that Ineffective<br>HAZOP study is identified as one of root causes or Ineffective management of HAZOP recommendation is identified as the one of root causes or is identified as the cause of insufficient protection layer. | 0 Case        |
| HAZOP-2/1: HAZOP Leader                      | Percent of HAZOP (MoC)                                                                                                                                                                                                                                                  | 100%          |

| KPI Measure                                                                 | Description / Calculation                                                                                                | Target (unit) |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------|
| qualification                                                               | performed by qualified and independent                                                                                   |               |
| HAZOP-2/2: HAZOP Leader qualification                                       | Percent of HAZOP (MoC) performed by disciplines and with experience as required by HAZOP study procedure                 | 100%          |
| HAZOP – 3/1: Non-registered of HAZOP Recommendation                         | Number of HAZOP study report that HAZOP Recommendations not registered in e-PHA for follow up                            | 0 item        |
| HAZOP – 3/2: Overdue HAZOP Recommendation                                   | Number of Active HAZOP Recommendation that is not "Closed out" and/or "Approved" within due date                         | 0 item        |
| HAZOP-3/3: Deviate closed out of HAZOP Recommendation without re-evaluation | Number of HAZOP Recommendation that is closed out deviate from original intent without reevaluation by HAZOP team member | 0 item        |
| HAZOP-3/4: Cancellation of HAZOP Recommendation without re-evaluation       | Number of HAZOP Recommendation that is cancelled without reevaluation by HAZOP team member                               | 0 item        |

### Related Law

| Law Name |
|----------|
|          |

### Related Documents

| Document ID        | Document Name                                           |
|--------------------|---------------------------------------------------------|
| P-(TP-PM)-OEMS-002 | Management of Change (MoC) Procedure                    |
| W-(Q-MP)-002       | Operational Risk Assessment Matrix and Its Applications |
| SG-(Q-MP)-004      | Guideline for Plant PSM Governance                      |
| SG-(Q-MP-PS)-001   | Process Safety Management Leading Indicators            |

**External Reference Documents**

| Document Name |
|---------------|
|               |

## Table of Contents

|    |                                     | Page |
|----|-------------------------------------|------|
| 1. | Purpose/Objective.....              | 1    |
| 2. | Scope .....                         | 2    |
| 3. | Roles and Responsibility .....      | 3    |
| 4. | Workflow.....                       | 9    |
| 5. | Detailed Narrative of Workflow..... | 10   |
| 6. | Appendix .....                      | 15   |

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

## 1. Purpose/Objective

**HAZOP** is a systematic methodology to identify hazards and operability problems which is the one of techniques for **Process Hazards Analysis (PHA)** as mention in the **Process Safety Management (PSM)** regulation of **Industrial Estate Authority of Thailand (IEAT)** and is generally accepted as the preferred methodology in the chemical and petroleum industries. The concept involves investigating how the plant might deviate from the design intent.

The aim of this procedure is to ensure the effective work process for HAZOP study in PTT Global Chemical Public Company Limited (hereinafter referred to as “**GC**”) by providing the requirement of role and responsibility and qualification of the person who are involved in HAZOP studied including work process for registration, approval of HAZOP report and follow-up process.

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

## 2. Scope

This procedure is applied to *HAZOP study* performed in *GC*, when *Change Review team (CRT)* of Plant change determine that *HAZOP study* is required as per P-(TP-PM)-OEMS-002 Management of Change Procedure

Internal Use Only

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

### 3. Roles and Responsibility

#### 3.1 Role & Responsibility

##### 3.1.1 HAZOP Procedure owner

HAZOP Procedure owner is Management System and Process Safety (Q-MP).

HAZOP Procedure owner shall have responsible tasks including

- Approve/Review the HAZOP study procedure when the document is revised and oversee effective of **HAZOP**.

##### 3.1.2 HAZOP Procedure custodian

HAZOP procedure custodian is staff of Technical Safety Services (Q-MP-TS).

HAZOP procedure custodian shall have responsible tasks including.

- Issue the HAZOP study procedure.
- Review and identify the opportunities for improving HAZOP study process.
- Provide HAZOP training courses.
- Certify and record the internal and external HAZOP Leader.
- Ensure that this procedure is implemented effectively and communicated to all relevant users.
- Perform the continuous and planned monitoring program
- Review opportunity for improvement of HAZOP study through
  - Feedback to key stakeholders.
  - Evaluation of areas of outstanding and improvement.
  - Providing recommendations.
  - Follow up gap closing

##### 3.1.3 HAZOP Coordinator

For HAZOP under scopr of **MoC**, HAZOP Coordinator is **MOC Champion** who has been assigned by Plant VP.

HAZOP Coordinator shall have responsible tasks including, but not limited to;

- Define the scope and objectives of the study.
- Planning HAZOP schedule and Coordinate the **HAZOP study**.
- Appoint a HAZOP Leader, HAZOP team members, and HAZOP scribe.
- Ensure that HAZOP Leader has the qualification as requirement.



- Prepare **Process Safety Information (PSI)** for HAZOP team and ensure that necessary information is distributed to HAZOP team members (at least 3 days prior to HAZOP Workshop start is preferred).
- Ensure that **(PSI)** is available and accurate for HAZOP study.
- Complete the PSI readiness checklist (Including P&ID readiness checklist) and send evaluation results to HAZOP Leader before conducting study.
- Arrange a HAZOP study.
- Ensure that all HAZOP recommendation items from HAZOP study are tracked and implemented properly.
- Raise the **HAZOP action change** and get agreement from HAZOP team when **HAZOP action change** is required.
- Ensure that the most updated **HAZOP study** report is registered (For HAZOP under scope of **MoC**, HAZOP study report shall register in the **e-PHA** system.)
- Monitor the HAZOP schedule and update to ensure that HAZOP workshop is completed as planned.

#### 3.1.4 HAZOP Leader

HAZOP Leader is **GC** staff or qualified contractor who has experience and is qualified as the requirement of this procedure.

HAZOP Leader shall have responsible tasks including, but not limited to;

- Review the PSI readiness checklist.
- Advising project/site leadership of the need to delay/postpone the study until issues affecting the integrity of the HAZOP can be resolved.
- Gather information for HAZOP node preparation for the **HAZOP study**.
- Review the design to determine sufficient technical information and required skills of study team members.
- Verify adequacy of team member to ensure effectiveness of the study meeting with multi-disciplinary team and using “brainstorms” approach.
- Facilitate the meeting/workshop and ensure that all issues are covered in adequate depth.
- Being alert to time pressures and ensure that it does not compromise the quality, thoroughness, or integrity of the review.
- Review and approve **HAZOP study** report.

#### 3.1.5 HAZOP Team members

HAZOP team members comprise of multi-discipline members who have experience and knowledge in their represented field.

HAZOP team members shall have responsible tasks including, but not limited to;

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

- Fully participate in the **HAZOP study** to their fullest ability and active (Core team member).
- Identify credible causes, develop consequences regarding hazards and operability problems including existing safeguard.
- Evaluate the risk level and suggest possible recommendations for prevention and mitigation measures to reduce the risk level.
- Review the **HAZOP action change** request.

### 3.1.6 HAZOP Scribe

HAZOP Scribe shall be the person who has a strong knowledge about HAZOP study. Each HAZOP study should have a dedicated HAZOP scribe which is not involved in brainstorming to increase the effective recording.

HAZOP scribe shall have responsible tasks including, but not limited to;

- Record identified hazards and problem areas and capable of structuring recommendations/ actions in a clear and understandable way.
- Complete the HAZOP report and send the final draft to HAZOP Leader for approval.

### 3.1.7 Responsible person

Responsible person shall be the person who is assigned by HAZOP team.

Responsible person shall have responsible tasks including, but not limited to

- Implement or/and follow up the action item and close action item as agreed by the HAZOP team. (For HAZOP under scope of **MoC**, action item shall be closed and recorded in the **e-PHA** system.)

### 3.1.8 Action item approver

Action item approver shall be the person who is assigned by HAZOP team. (For HAZOP under scope of MoC, if there is no specific person as team consensus, **MoC champion** shall be the action approver.)

Action items approver shall have responsible tasks including, but not limited to

- Action item approver is the person who has been assigned to review, approve or reject proposed closed-out action item. If there is no specific person as team consensus, **MoC champion** shall be the action approver.

### 3.1.9 PHA Element leader

PHA element leader is the Division Manager of Plant technic.



PHA element leader or Person who assigned by PHA element leader shall have responsible tasks including, but not limited to

- Review results of monitoring program and communicate to Plant PSM committee include request the resource via Plant PSM committee as required.
- Monitoring HAZOP recommendations overdue as requirement of SG-(Q-MP-PS)-001 Process Safety Management Leading Indicators and communicate to Plant PSM committee include request the resource via Plant PSM committee as required.

Internal Use Only

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

### 3.2 Table 1: Qualifications of HAZOP Staffs

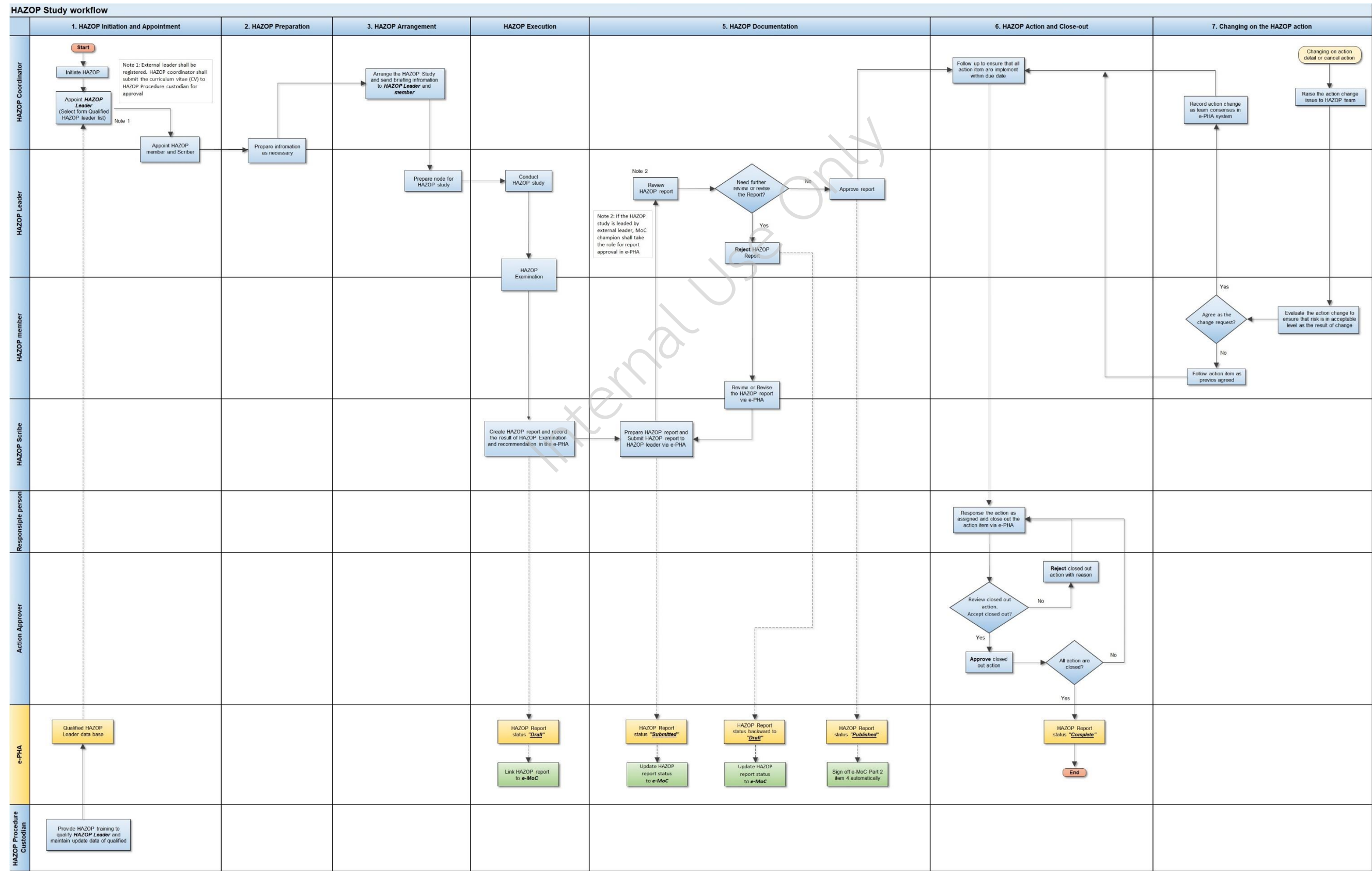
| Role                     | Qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Responsible party                                                                         | Qualified by              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------|
| <b>HAZOP Coordinator</b> | <ul style="list-style-type: none"> <li>Qualified <b>MoC champion</b> as P-(TP-PM)-OEMS-002 Management of Change Procedure.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li><b>MoC champion</b></li> </ul>                     | TP-PM                     |
| <b>HAZOP Leader</b>      | <p>Internal HAZOP Leader</p> <ul style="list-style-type: none"> <li>Minimum 8 years' experience in Refinery, Petrochemical, or Chemical process.</li> <li>Trained HAZOP Leadership course.</li> <li>Trained Awareness of Process Safety in Design.</li> <li>Must be independent person.</li> </ul> <p><i>Note Independent means;</i></p> <p><b>Inthernal HAZOP Leader:</b> HAZOP leader independent from project organization, EPC contractor/subsidiary of MoC being reviewed and independent of line of management in VP Operation level (Plant VP)</p> <p><b>External HAZOP Leader:</b> HAZOP leader independent from project organization, EPC contractor/subsidiary of MoC being reviewed</p> <p>External HAZOP Leader</p> <ul style="list-style-type: none"> <li>External leader shall be qualified and registered by HAZOP Procedure custodian.</li> </ul> <p><b>Note</b> HAZOP coordinator shall submit the curriculum vitae (CV) to HAZOP Procedure custodian for approval. Qualification of external leader are provided in <b>HAZOP Study guideline</b></p> | <ul style="list-style-type: none"> <li><b>GC</b> staff or qualified contractor</li> </ul> | HAZOP Procedure Custodian |
| <b>HAZOP Scribe</b>      | <ul style="list-style-type: none"> <li>Minimum 2 years' experience in Refinery, Petrochemical, or Chemical process.</li> <li>Should be trained or familiar with <b>HAZOP study</b> methodology.</li> </ul> <p>Internal resources should be trained in an <b>e-PHA</b> system</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li><b>GC staff</b> or qualified contractor</li> </ul> | HAZOP Procedure Custodian |

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

| Role                             | Qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Responsible party                                                                                                                                         | Qualified by              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>HAZOP team</b>                | <p>Core team members (participate full time)</p> <ul style="list-style-type: none"> <li>Required disciplines of core team members are as; <ol style="list-style-type: none"> <li><b>MOC champion</b></li> <li>Operation representative<sup>Note</sup></li> <li>Process engineers representative<sup>Note</sup></li> </ol> <p><i><b>Note</b> Minimum 3 years' experience in Refinery, Petrochemical, or Chemical processes, otherwise he/she shall be assigned by Division manager to participate in the <b>HAZOP study</b></i></p> <ol style="list-style-type: none"> <li>Site Maintenance Engineer</li> <li><b>Plant SHE</b> representative</li> </ol> </li> <li>Should be trained or familiar with <b>HAZOP study</b> methodology.</li> </ul> <p>Specialist may be additionally required by <b>HAZOP Leader</b>. For example, Process Control engineer, Rotating specialist etc.</p> | <ul style="list-style-type: none"> <li><b>Change Review Team</b></li> <li>Multi-discipline members as required (<b>GC staff</b> or contractor)</li> </ul> | HAZOP Procedure Custodian |
| <b>Responsible person</b>        | <ul style="list-style-type: none"> <li>Discipline Engineer or the Person who has knowledge related to action content.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li><b>GC staff</b> or contractor</li> </ul>                                                                           | -                         |
| <b>Action approver</b>           | <ul style="list-style-type: none"> <li>Assigned as consensus of HAZOP team. If there is no specific person as team consensus, MoC champion shall be the action approver</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li><b>GC staff</b></li> </ul>                                                                                         |                           |
| <b>HAZOP Procedure Custodian</b> | <ul style="list-style-type: none"> <li>Q-MP-TS staffs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Q-MP-TS</li> </ul>                                                                                                 |                           |



4. Workflow



|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

## 5. Detailed Narrative of Workflow

### 5.1 HAZOP Initiation and Appointment

- *HAZOP study* is a part of *Management of change procedure*.
- *HAZOP study* will be initiated when *Change Review Team* has determined that *HAZOP study* is required as the result of the *PRELIMINARY, SHE ASSESSMENT AND SHE EVALUATION*.
- HAZOP Coordinator makes appointments with HAZOP Leader, HAZOP scribe and HAZOP team members

Note SG-(Q-MP)-011 Guidance for Hazard and Operability Studies (HAZOP) shall be applied if the HAZOP Term of reference (TOR) is required.

### 5.2 HAZOP Study preparation

- HAZOP Coordinator defines scope of *HAZOP study* with assistance by HAZOP Leader and provides the necessary information and resources for study.
- HAZOP Leader prepares *HAZOP study* plan which shall contain the general information follows.
  - Objective and scope of the study
  - List of participating members.
  - Defined nodes, systems design intent.
  - List of proposed guide words to be used.
  - Risk Assessment Matrix.
  - HAZOP worksheet
- HAZOP Coordinator shall prepare technical information before study begins, the following documentation which shall be clearly and uniquely identified, approved and dated must be provided (Recommend 3 days in advance).
  - a) Design requirement and description
  - b) SDS (Safety Datasheet),
  - c) Plot Plan & Layout,
  - d) PFD (Process Flow Diagram),
  - e) P&ID (Piping and Instrumentation Diagram),
  - f) Control & Safeguarding Diagram,
  - g) Alarm and Trip Setting,
  - h) Piping class specification,
  - i) Equipment datasheet,
  - j) Relief and blowdown summary table,
  - k) Operating and maintenance procedures such as start-up, shutdown and emergency, etc. as applicable.

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

- HAZOP Coordinator (or Project team) should conduct PSI readiness checklist to ensure that the accuracy and adequacy of PSI is sufficient for HAZOP study. Refer Appendix 6.2 (Table A6.2-2 and Table A6.2-3) in SG-(Q-MP)-014 Guidance for Hazard and Operability Studies (HAZOP). Evaluation result shall be informed to HAZOP Leader before HAZOP study session starts (At least 3 days before HAZOP study session is recommended).

### 5.3 HAZOP Arrangement

- HAZOP Coordinator arranges a **HAZOP study** and sends out the invitation with the following information to HAZOP team.
  - a) HAZOP schedule and venue.
  - b) Technical information required for the **HAZOP study**.
- HAZOP Coordinator should send the necessary information to HAZOP team members to familiarize them with system being reviewed. The necessary information includes, but not limited to, the design requirement and process description, plot plan & Layout, PFD (Process Flow Diagram), etc.

### 5.4 HAZOP Study

- During a brainstorming of the **HAZOP study**, the following steps shall be followed
  - a) At the beginning, HAZOP Leader shall give presentation to the team through the following list:
    - Introduce team members
    - Sign a HAZOP attendance sheet
    - Brief **HAZOP study** methodology including Risk Assessment Matrix (RAM)
    - Present “study ground rules”
  - b) HAZOP leader then starts **HAZOP study** by selecting a study node and explains node boundary to team members.
  - c) Process engineers shall then provide design intent of the node included operating conditions to the team members.
  - d) Then HAZOP Leader chooses Deviation, to stimulate team members to identify all credible cause(s) using brainstorming technique.
  - e) Team members shall develop possible consequence(s) and determine initial risk using risk matrix. At this stage, severity and likelihood shall be determined without any safeguards in place.

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

- f) After that the HAZOP team shall define all existing safeguard(s) and evaluates mitigated risk by re-evaluation of likelihood by takes into account the existing safeguards.

Note: Severity shall be remained unchanged as the initial risk evaluation

- g) If mitigated risk is unacceptable, recommendation(s)/action(s) shall be provided to reduce risk to an acceptable level. However, recommendation/action can be suggested based on team's consensus even if the mitigated risk is in acceptable level in order to improve plant operability.
- h) Repeat step d) through g) until all applicable deviations are considered.
- i) Repeat step b) through h) until all nodes are studied.

**Note** Guidance practice of HAZOP methodology as provided in section 5.3.3 of SG-(Q-MP)-014: Guidance for Hazard and Operability Studies (HAZOP) shall be applied.

- HAZOP scribe shall record HAZOP minutes in precise and concise in the HAZOP worksheet.

**Note** HAZOP report status in **e-PHA** shows "**draft**". When the PHA no, is created in **e-PHA**

- HAZOP coordinator shall ensure that all action items are assigned to a responsible person to close-out action or tracking and follow-up. It includes Action approver for approval of proposed action closed out. Action items approver may be identified as team consensus.

If there is no specific responsible person is assigned, **MoC champion** shall take responsibility as the Action Approver.

**Note** For the action item that responds by an external person, the Company name of the contactor shall be recorded in the worksheet. In this case, **MoC champion** shall be the responsible person to track follow-up actions to be closed out in a timely fashion.

## 5.5 HAZOP Documentation

- HAZOP scribe prepares the draft of HAZOP report and submits to HAZOP Leader to review via **e-PHA** system.
- When HAZOP study is carried out by an external HAZOP leader, HAZOP coordinator shall register the HAZOP report is **e-PHA** as the attached file and register HAZOP action item in the **e-PHA** for the follow-up process

**Note** HAZOP report status in **e-PHA** shows "**submitted**" after the report is submitted to HAZOP leader

- HAZOP Leader reviews and approves the HAZOP report via **e-PHA** in case no further comment. HAZOP leader can reject the report if additional review is required

**Note** HAZOP report status in **e-PHA** show "**Publish**", if the report is approved

**Note** HAZOP report status in **e-PHA** show "**Draft**", if the report is rejected.

**Note** If the **HAZOP study** is led by an external leader, HAZOP Coordinator shall take the role for report approver instead HAZOP Leader to approve the report in **e-PHA** system

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

- HAZOP coordinator shall ensure that the HAZOP report is linked to the *e-MoC*

## 5.6 HAZOP Action and Close-out

- HAZOP Coordinator shall distribute/communicate the actions (action plan, action to be taken, completed date, and responsible person) from *HAZOP study* to the responsible person.
- All recommendations shall be closed out by the responsible person and approve by *MoC champion* or the person who is assigned by *MoC champion* via *e-PHA*.
- HAZOP Coordinator shall follow-up actions from *HAZOP study* and ensure that all action item is closed out by responsible person within the due date

*Note If all actions are closed out, the HAZOP report status in e-PHA shows "Complete".*

## 5.7 HAZOP Action change

- During action closing-out, the Responsible person, HAZOP coordinator, or any related person could propose a resolution which may differ from the original recommendation, if the original HAZOP recommendation is not possible / not practical to implement or the newly proposed action will provide a safer solution. In such cases, the *HAZOP action change* process as described below shall be followed
  - a) Proposed change shall be informed by HAZOP coordinator to evaluate the necessities of change.
  - b) HAZOP coordinator shall ensure that risk level of all hazardous scenario relates to propose changed HAZOP recommendations are reviewed and is in acceptable level. Re-evaluation shall be performed by HAZOP team (Original HAZOP member should be involved)
  - c) HAZOP coordinator shall record action change in *e-PHA* as team consensus, if HAZOP team agreed that action change is required.

## 5.8 Monitoring Program

- The HAZOP audit process is a regular basis that contributes to management control to ensure compliance, quality of study and recommendation management by the following audit program.

### a) Monthly monitoring

- PHA Element Leader or Assigned staff by PHA Element Leader shall monitor the overdue HAZOP recommendation followed the criteria of SG-(Q-MP-PS)-001: Process Safety Management Leading Indicators.

Note Below are the criteria of HAZOP recommendation overdue for Monthly monitoring follow SG-(Q-MP-PS)-001: Process Safety Management Leading Indicators.

|                   |                                                                                                                                                                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indicator         | No HAZOP Recommendation Overdue                                                                                                                                                                                                                                                                                  |
| Target            | 0 Recommendation                                                                                                                                                                                                                                                                                                 |
| Detail            | <p>No “Active” HAZOP Recommendation pending in status of overdue during the time of tacking.</p> <p>Active HAZOP recommendation mean HAZOP Recommendation that not yet closed out by Responsible person or is closed out but in awaiting approval by Approver (In e-PHA dash show in “In-progress” category)</p> |
| Period /Frequency | 1 time per month.                                                                                                                                                                                                                                                                                                |
| Approach          | Tracking from Dashboard in e-PHA system                                                                                                                                                                                                                                                                          |

- The monthly monitoring result shall be reported to PHA element leader.
- PHA Element leader has responsibility to report in Plant PSM committee.

b) Continuous Monitoring (PSE Tier 1 and Tire 2 relate to HAZOP Study)

- HAZOP Procedure custodian has responsibility for tracking PSE Tier 1 and Tire 2 and evaluating the PSE detail against criteria of HAZOP Procedure KPI 1, See 6.4.
- HAZOP Procedure custodian shall inform the result of evaluation to PHA Element Leader of plant relate to PSE being evaluated when there is case meet the criteria of HAZOP Procedure KPI 1.
- PHA Element Leader has responsibility to review the evaluation result obtained from HAZOP Procedure custodian and finalize the evaluation result including preventive and/or corrective action together with HAZOP Procedure custodian before reporting to Plant PSM committee.
- PHA Element Leader has responsibility to report the evaluation result including preventive or corrective action to Plant PSM committee.

c) Planned Monitoring (Corporate Audit)

- HAZOP Procedure custodian has responsibility to perform planned monitoring (as plan) for HAZOP Procedure KPI 2/1, 2/2, 3/1, 3/2, 3/3, 3/4, See 6.4.
- HAZOP Procedure custodian shall report results to PHA Element Leader and finalize Finding, Preventive and/or Corrective action together with PHA Element Leader.
- PHA Element Leader has responsibility to report the evaluation result including preventive or corrective action to Plant PSM committee.

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

## 6. Appendix

### 6.1 Terms and Definitions

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i><b>HAZOP Action change</b></i>   | The changing in detail of action or action cancellation which deviate from the original approved report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <i><b>HAZOP Study</b></i>           | A structured and systematic technique for examining a defined system, to identify potential hazards and operability problems in the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <i><b>HAZOP Examination</b></i>     | The proceeds by systematically using a series of <i><b>guide words</b></i> to identify the potential <i><b>deviations</b></i> from the <i><b>design intent</b></i> and employing these <i><b>deviations</b></i> as “triggering devices” to stimulate the team members to envisage how the <i><b>deviation</b></i> might occur and what might be the <i><b>consequences</b></i> . The examination relies on specialists from various disciplines with appropriate skills and experiences who display intuition and good judgment. The <i><b>HAZOP</b></i> examination shall be carried out in a climate of positive thinking and frank discussion. Solutions to identify the problems are not the primary objective of the <i><b>HAZOP</b></i> examination, but if made they will be recorded for consideration by those responsible for the design. |
| <i><b>Possible Cause</b></i>        | The reasons why <i><b>deviations</b></i> might occur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <i><b>Potential Consequence</b></i> | The hazardous event that might occur as the result of <i><b>deviation</b></i> without consideration of existing safeguard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <i><b>Change</b></i>                | Any action or work which results in permanent or temporary <i><b>Change</b></i> to the physical facilities, operating conditions, operating procedures, or organizational structure that may affect the safety, health or environment of company or contract personnel, and/or the community.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <i><b>Design intent</b></i>         | Designer’s desired, or specified range of behavior for parameters and characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <i><b>Deviation</b></i>             | The combination of parameter and guideword that help the HAZOP team investigate how the plant might <i><b>deviate</b></i> from the <i><b>design intent</b></i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <i><b>Guideword</b></i>             | Word or phrase which expresses and defines a specific type of <i><b>deviation</b></i> from a <i><b>parameter</b></i> ’s <i><b>design intent</b></i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <i><b>Harm</b></i>                  | Physical injury or damage to the health of people or damage to property or the environment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

### **Hazard**

A physical situation, condition or material property that has the potential to cause harm such as sickness, injury or death to people, damage to property and investments, environmental impact, business interruption and loss of reputation. For example - crude oil is a dangerous substance which can cause a lot of harm when treated without case

### **Change Review Team**

**Change Review Team** is a multi-disciplinary team. It is typically composed of, but not limited to, the people participating in **BDRs** (**BDR**) at least **MOC Champion**, **Plant VP** representative (normally is the representative from Operation division) and **SHE** representative, **Process Engineer** and **Site Maintenance Engineer** and/or **Plant Asset** and disciplines that might be affected by the **Change**. (Refer to P-(TP-PM)-OEMS-002: Management of Change)

**Change Review Team** may also include a **project** engineer, rotating engineer, inspection engineer, I&E engineer, process control engineer, supply chain management representative, specialist, or any other required party deemed appropriate.

**Change Review Team** are qualified by minimum 3 years' experience with training background of **Change Review Team**, basic HAZOP or PHA (Process Hazard Analysis) courses.

For all **Changes** a representative from Operation Division is required to attend. They are the ultimate owners of the **Change** and are therefore critical to ensuring that the **Change** meets operational requirements and can be successfully and safely operated.

### **Initial risk**

The risk ranking without consideration of the existing safeguard

### **Likelihood**

A measure of the expected probability or occurrence frequency of unmitigated consequence

### **Line Manager**

Informant's Division Manager

### **Mitigated risk**

The risk ranking with consideration of the existing safeguard

### **MOC**

Management of Plant Change (**MOC**) means management process for Changes which related to all Changes in Production Units.

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

### *MOC Champion*

Anyone who is assigned to execute the *Change*. *MOC Champion* takes *MOC* through the entire process, ensuring that all appropriate reviews, procedures and documentation are completed during the various stages of the *MOC*. *MOC Champion* shall ensure that all the appropriate approvals are obtained for Implementation, Start-up and Close-out of the *Change*.

### *Node*

A specific location in the process in which (the deviations of) the design/process intent are evaluated. It may be physical (e.g. hardware) or logical (e.g. step in an operational sequence).

### *Parameter*

Variable that could *cause deviations* e.g. flow, pressure, temperature etc.

### *Plant Change*

The *Change* refers to any modifications or *Changes*, either temporary or permanent ones, of any equipment, piping system, distribution system, civil foundation system including equipment structure of operational process, fire protection and firefighting system that impact to operational process and any operating condition *Changes* those deviate from operation manual either within normal or abnormal operation which impact to process parameter such as flow rate, temperature, pressure or other chemical or physical properties.

### *GC*

PTT Global Chemical Public Co., Ltd. and subsidiaries including GC Group.

### *Risk*

Combination of the probability of occurrence of *harm* and the *severity* of that *harm*

### *Risk Assessment Matrix*

A matrix that is used during *risk* assessment to define the various levels of *risk* as the product of the *harm severity* categories and *harm likelihood* categories

### *Existing Safeguard*

A facility, process or procedure that helps to reduce the *likelihood* or frequency of the *deviation*, or to mitigate its *consequence*

### *Severity*

A measure of the expected impact of hazardous event

### *Plant SHE*

Staff in SHE area services division that provides Safety, Health and Environment Service for each responsible plant area

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

## 6.2 Other Supporting Information

### 6.2.1 HAZOP Study form

- HAZOP template shall provide the content comply with the required content as mentioned in SG-(Q-MP)-014 Guidance for Hazard and Operability Studies (HAZOP)

Note *HAZOP study* worksheet provides the *e-PHA* system

### 6.2.2 HAZOP Deviation

- Typical HAZOP deviation is provided in SG-(Q-MP)-014 Guidance for Hazard and Operability Studies (HAZOP)

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

### 6.3 Implementation Plan

- After this procedure is effective, HAZOP report shall be registered via **e-PHA** system due to PHA numbering will link to e-MoC numbering and allow automatic sign off e-MoC part 2 item 4 when HAZOP report is approved in **e-PHA** (for the **e-MoC no** that was created before **e-PHA** Go-Live,1/02/2020).
- The following are amendment detailed narrative of workflow for section 5.5 which shall be applied for **e-MoC** that was created before **e-PHA** Go-Live (3/03/2020).
  - HAZOP scribe prepares the draft of HAZOP report and submits to HAZOP Leader to review via **e-PHA** system.  
*Note* HAZOP report status in **e-PHA** shows “**submitted**”
  - HAZOP Leader reviews and approves the HAZOP report via **e-PHA** if no further comment or reject to resubmit, if additional review is required.
  - For the outsource HAZOP report, HAZOP coordinator shall register the HAZOP report is **e-PHA** as the attached file and register HAZOP action item in the **e-PHA** for the follow-up process.  
*Note* HAZOP report status in **e-PHA** show “Publish”, if the report is approved.  
*Note* HAZOP report status in **e-PHA** show “Draft”, if the report is rejected.  
*Note* If the **HAZOP study** is led by an external leader, HAZOP Coordinator shall take the role for approval of **HAZOP study** report and Changing of action item in the **e-PHA** system.
  - HAZOP coordinator shall attach HAZOP report manually in **e-MoC** part 2 item 4 after HAZOP report in **e-PHA** system is approved by HAZOP Leader and shown status “Publish”.
- After the procedure effective, the HAZOP methodology as mentioned in SG-(Q-MP)-014 Guidance for Hazard and Operability Studies (HAZOP) should be applied to perform HAZOP study as the good practice during ranking the mitigated risk.
- This procedure was revised to add details of monitoring program and was approved for implementation in 2025. The monitoring program shall be implemented in 2026 after procedure is registered in Document Control System (DCM)

#### 6.4 HAZOP KPI

##### HAZOP – 1: PSE Tier 1, 2 relate to HAZOP study

**Target:** 0 Case

##### **Description**

Number of PSE (Tier 1 and Tier 2) for MoC that;

1. Ineffective HAZOP study is identified as one of root causes.

“Ineffective HAZOP study” mean

- Facility/Unit/Equipment that incident occurs are not included in the HAZOP study.
- Operation mode (in normal operation) relating to incident events is not reviewed.
- HAZOP Deviation relates to incident events is not reviewed.
- Scenarios relating to incidents leave an unacceptable risk without sufficient protection barrier to achieve acceptable risk.
- Improper/Incorrect PSI leading to miss-identification of hazard scenario, mission/incorrect identification of possible cause, incorrection potential consequence, incorrect safeguard identification.

2. Ineffective management of HAZOP recommendation is identified as the one of root causes or is identified as the cause of insufficient protection layer.

“Ineffective management of HAZOP recommendation” mean

- HAZOP recommendation relating to incident event is planned to complete in improper MoC part lead to insufficient/lost protection barrier.
- HAZOP recommendations relate to incident events are not registered which this deviation is identified as one of incident root causes or insufficient protection layer.
- HAZOP recommendation relates to incident event is not closed out within due date which this deviation identified as one of root causes or insufficient protection layer.

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

- HAZOP recommendation relates to incident event is closed out deviate from original recommendation intent or is cancelled without reevaluation by HAZOP team.

#### **Approach for monitoring**

- Tacking PSE Tier 1 and Tier 2 from incident investigation system when Tier is confirmed.
- Evaluate based on criteria of “Ineffective HAZOP study” and “Ineffective management of HAZOP recommendation”.
- Only HAZOP study under MoC is included (HAZOP study in scope of HAZOP Revalidation or Original HAZOP are excluded).

#### **HAZOP – 2/1: HAZOP Leader qualification**

**Target:** 100%

#### **Description**

Percent of HAZOP (MoC) performed by qualified and independent  
“Independent” mean

- Independent form Project organization
- Independent Line of management in VP Operation level (Plant VP)
- Independent form EPC contractor/subsidiary of MoC being reviewed

#### **Approach for monitoring**

- Sampling report from e-PHA system within the interval (typically 2 year) between year of previous sampling and current year.

Note Number of samplings shall be 50% of total number of HAZOP report in sampling period but not less than 5 report. All of report shall be sampling if total number of reports is less than 5 reports in sampling period.

#### **HAZOP – 2/2: HAZOP Member qualification**

**Target:** 100%

#### **Description**

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

Percent of HAZOP (MoC) performed by disciplines and with experience as required by HAZOP study procedure.

- MoC Champion (Project representative for plant change by plant project)
- Operation representative (3 years or assigned by DM)
- Process engineer (3 years or assigned by DM)
- Site maintenance representative
- Plant Safety engineer

#### **Approach for monitoring**

- Sampling report from e-PHA system within the interval (typically 2 year) between year of previous sampling and current year.

Note Number of samplings shall be 50% of total number of HAZOP report in sampling period but not less than 5 report. All of report shall be sampling if total number of reports is less than 5 reports in sampling period.

#### **HAZOP – 3/1: Non-registered of HAZOP Recommendation**

**Target:** 0 non-registered HAZOP recommendation

#### **Description**

Number of HAZOP study report that HAZOP Recommendations not registered in e-PHA for follow up.

#### **Approach for monitoring**

- Tracking list of MoC report with part 2.4 (In-depth SHE assessment) from e-MoC system and Tracking registration of HAZOP report and recommendation in e-PHA system.

#### **HAZOP – 3/2: Overdue HAZOP Recommendation**

**Target:** 0 overdue HAZOP recommendation

#### **Description**

Number of Active HAZOP Recommendation that is not "Closed out" and/or "Approved" within due date.

#### **Approach for monitoring**

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

- Sampling report from e-PHA system within the interval (typically 2 years) between year of previous sampling and current year.

Note Number of samplings shall be 50% of total number of HAZOP report in sampling period but not less than 5 report. All of report shall be sampling if total number of reports is less than 5 reports in sampling period.

### **HAZOP – 3/3: Deviate closed out of HAZOP Recommendation without re-evaluation**

**Target:** 0 Deviated closed out of HAZOP recommendation without re-evaluation and/or risk is not demonstrated to acceptable level.

#### **Description**

Number of HAZOP Recommendation that is closed out deviate from original intent without reevaluation by HAZOP team member.

#### **Approach for monitoring**

- Sampling report from e-PHA system within the interval (typically 2 years) between year of previous sampling and current year.

Note Number of samplings shall be 50% of total number of HAZOP report in sampling period but not less than 5 report. All of report shall be sampling, if total number of reports less than 5 report in sampling period.

### **HAZOP – 3/4: Cancellation of HAZOP Recommendation without re-evaluation**

**Target:** 0 Cancellation of HAZOP recommendation without re-evaluation and/or risk is not demonstrated to acceptable level.

#### **Description**

Number of HAZOP Recommendation that is cancelled without reevaluation by HAZOP team member.

#### **Approach for monitoring**

- Sampling report from e-PHA system within the interval (typically 2 years) between year of previous sampling and current year.

|                                                                                   |                                               |                                                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
|  | PTT Global Chemical Public<br>Company Limited | P-(Q-MP)-OEMS-005: Hazard and Operability<br>Study (HAZOP) |
|-----------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|

Note Number of samplings shall be 50% of total number of HAZOP report in sampling period but not less than 5 report. All of report shall be sampling if total number of reports is less than 5 reports in sampling period.

Internal Use Only